

Closeness and Strategic Participation: Does the Relative Closeness of the US Presidential Elections Shape Where College Students Register to Vote?

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Existing research shows that participation rates are higher in competitive electoral contexts. However, it is difficult to disentangle whether this contextual effect is a function of geographic heterogeneity or the strategic incentives for participation that individuals face in close elections. In this article, we partially circumvent this dilemma by focusing on out-of-state college students facing the choice of where to cast their vote. Using a novel survey and administrative records for over 1 million out-of-state students, we show that individuals attending college in a state where the election is more competitive than in their home state are more likely to register in their campus state, consistent with theories of strategic behavior. We argue that by focusing on this population and this question we are better able to assess the relationship between strategic incentives and political behavior while holding constant state-specific context.

Research on participation has consistently shown that individuals' context affects whether and how they engage politically (e.g., Huckfeldt, Plutzer, and Sprague 1993). In particular, scholars have shown that one contextual factor—election *competitiveness*—can affect rates of participation (Gimpel, Kaufmann, and Person-Merkowitz 2007; Pacheco 2008) and even how individuals respond to campaign activity (Fraga and Hersh 2010; Holbein and Hillygus 2016). However, although the relationship between competitiveness and participation rates is well established, its interpretation remains ambiguous. Consider turnout rates in US presidential elections. Is turnout higher in competitive states because of geographic heterogeneity, the electoral context (e.g., elite mobilization), or strategic calculus by voters?

We try to circumvent this ambiguity by examining students attending out-of-state colleges during presidential elections to determine whether they register to vote in their home states or

in the states where they attend college. Our theory is that some students will be strategic when deciding where to register. Specifically, *we hypothesize that students are more likely to register in their campus state (relative to their home state) if the election in their campus state is seen as more competitive.*

There are several advantages to focusing on this population and outcome. First, out-of-state college students (OSCS) represent a large group facing differing relative contextual incentives while holding state-specific factors constant. The set of presidential candidates remains fixed as does the importance, content, and closeness of the national race. In addition, state-specific electoral context (e.g., campaign activity) is the same for all students attending college together. This allows us to use standard tools to statistically isolate the effect of state-dyadic contextual factors (the difference in competitiveness) while holding constant the specific context of students' campus state, home state, or (to a lesser extent) both. Moreover, college

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Replication files are available in the JOP Dataverse (<https://dataverse.harvard.edu/dataverse/jop>). The empirical analysis has been successfully replicated by the JOP replication analyst. An online appendix with supplementary material is available at <https://doi.org/10.1086/716287>.

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Table 1. Measuring Relative Election Closeness (REC) for Two Students, 2016

	Home State	Campus State	REC
Case 1	Alabama	Florida	
	$ 62.1^* - 34.4^\dagger = 27.7$	$ 49.0^* - 47.8^\dagger = 1.2$	$27.7 - 1.2 = 26.5$
Case 2	Alabama	Mississippi	
	$ 62.1^* - 34.4^\dagger = 27.7$	$ 57.9^* - 40.1^\dagger = 17.8$	$27.7 - 17.8 = 9.9$

* Republicans' expected vote share in the 2016 election.

† Democrats' expected vote share in the 2016 election.

attendance patterns among OSCS are driven by nonpolitical factors such that relative competitiveness is orthogonal to standard predictors of participation (e.g., political interest) at the individual level.¹

Thus, this design allows us to more plausibly isolate the role strategic considerations play in shaping behavior while accounting for individual-level and state-level confounders. We cannot entirely rule out state-dyadic-level confounding, but below we take several additional steps to reduce this probability, including controlling for dyadic-level factors, estimating campus-state and home-state random effects simultaneously, and directly assessing the role of strategic considerations in registration decisions in a small survey of OSCS.

Second, in examining this question we can leverage the power of administrative records to bypass concerns about the accuracy of survey self-reports. Although we do rely on novel surveys of college students below, our main analysis is based on the remarkable National Study of Learning, Voting, and Engagement (NSLVE) for 2012 and 2016. The NSLVE uniquely combines university administrative records with voter files to identify where students were registered at the time of the election. Thus, we are able to test our theory with data from over 1 million politically active OSCS from over 1,000 post-secondary institutions.

Finally, the ability of students to register in their campus states has become a matter for policy debate driven by fears of OSCS voting in large numbers. Several states have taken steps to prevent them from registering locally, including refusing to accept student IDs and eliminating campus voting sites (Specht 2018). Indeed, in 2018 New Hampshire passed a law that critics claim was designed to prevent OSCS registration (Wilson 2018). Some politicians have gone so far as to allege widespread voter fraud by OSCS (Doran 2018), suggesting that this policy debate will continue and that further investigation is warranted.

To our knowledge, Niemi and Hanmer (2010, 314) offer the only previous study to explore the link between competitiveness and registration location. Using a survey of college students, they cross-tabulate registration decisions with a dichotomous indicator of whether students' home states or campus states are "battlegrounds." However, this was not the focus of the study, and the sample included only 90 OSCS facing a strategic decision.²

Before turning to our analysis, we first define relative election closeness, our measure of relative competitiveness. To make this concrete, we provide two examples in table 1. The first (case 1) is a student from Alabama who goes to neighboring Florida for college. The difference in the closeness of the race in those two states works out to 26.5. This is in contrast to an Alabama resident who goes to college in Mississippi (case 2). Since both states are fairly noncompetitive, the difference in closeness is 9.9. Our theory is simply that a student from Alabama who goes to college in Florida is more likely to register in Florida than a student from Alabama who goes to college in Mississippi is likely to register in Mississippi because Florida is a strategic target while Mississippi and Alabama are not.

NORTH CAROLINA STUDENT SURVEY

Our first study is a survey of 960 OSCS at one university in North Carolina in 2008. The survey, conducted October 27–30, included questions about students' state of residence as well as the state in which they would choose to register their vote. The outcome is a binary variable coded 1 if students indicated that they wished to register in North Carolina (54%).³

We calculate *relative election closeness* (REC) using the expected winning margin in each state obtained from the pre-election state-level forecasts from fivethirtyeight.com, a website

1. Appendix F.1 (apps. A–I are available online) tests this explicitly, using the student survey described below.

2. See app. G for a direct comparison of our results with Niemi and Hanmer (2010) and full replications of our results below using their "battleground" approach.

3. Appendix A includes response rates, survey questions, and descriptive statistics.

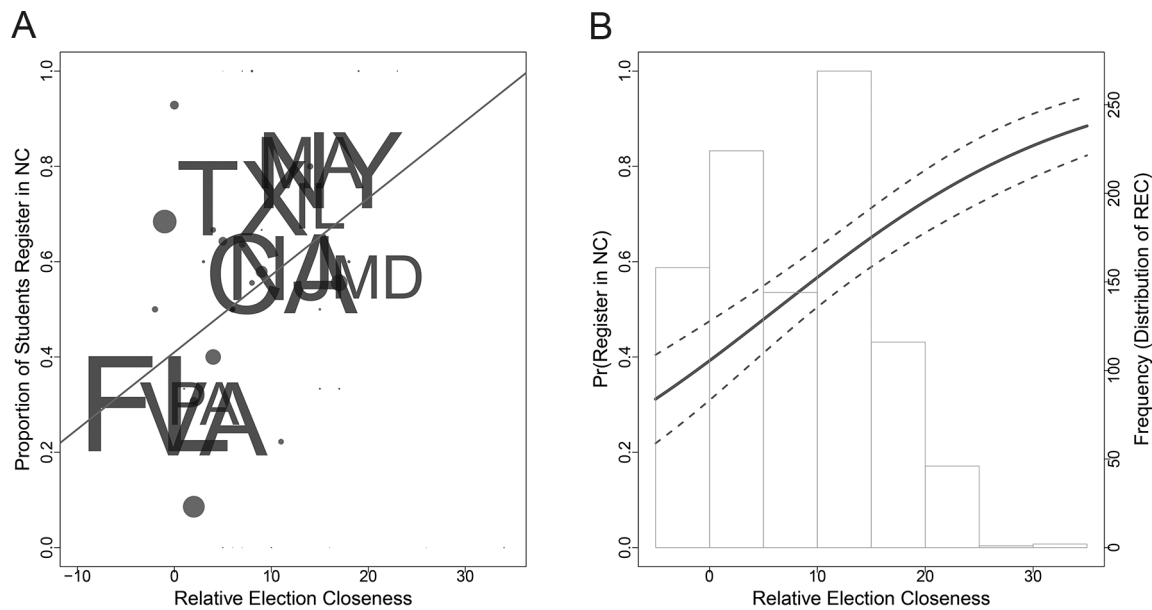


Figure 1. Relationship between relative election closeness (REC) and registration in North Carolina, 2008. *A*, Text/point size indicates the number of students from each state. State abbreviations are shown for only the largest states. The line represents a bivariate regression between REC and registration in North Carolina. *B*, Predicted value and 95% confidence interval of registering in North Carolina as a function of REC from a multivariate logistic regression. The histogram shows the distribution of REC.

led by Nate Silver. We argue that these forecasts represent something of a consensus about the state of the election since, first, it is based primarily on publicly available polling data and, second, the forecasts themselves were widely discussed.⁴ However, we show in appendix H that our results hold using the actual election results. Positive values for REC indicate that North Carolina was the more competitive state; thus, our theory is that there should be a positive relationship between REC and the decision to register in North Carolina.

Figure 1A shows the bivariate relationship between registration choice and REC, where students are grouped by home state. The size of the labeled states corresponds to the number of students. The line is the simple bivariate relationship as estimated with ordinary least squares (estimated at the individual level). The plot clearly shows that students from less competitive states (e.g., NY, IL, MA) are more likely to register in their campus state than students from highly competitive states (e.g., FL, PA, and VA). Moreover, these differences are large; around 30% of voters in Florida and Pennsylvania indicated they would register in North Carolina compared to nearly 80% of students from New York and Massachusetts.

Next we fit a logistic regression including controls for standard predictors of political participation including demograph-

ics (gender, race, and ethnicity), political characteristics (party identification), and home-state characteristics (adjacency to North Carolina, primary timing, and registration difficulty).⁵ The coefficient for REC is 0.071 (SE = 0.012, $p < .001$), supporting our hypothesis. The full model results are shown in appendix B, but in figure 1B we show the predicted probabilities based on a respondent whose covariates represent the average (or modal) characteristics of our sample. The probability of registering in North Carolina is roughly 39% for students from states similarly competitive to North Carolina and rises to 89% for students coming from the least competitive states. The average marginal effect (Hanmer and Kalkan 2013) of a 2 standard deviation change in REC is 22.8% (see app. B.2).

NSLVE ANALYSIS

The NSLVE combines administrative records from more than 1,000 postsecondary institutions and national voter files maintained by Catalyst for the 2012 and 2016 elections. Any institution can join while protecting student privacy, and around 80% are public two-year and four-year schools (we exclude community colleges and technical schools). The NSLVE includes student zip codes of residence, gender, age, race, voting eligibility, and institution identifier. It also includes voting

4. We use the forecast from right before the election (November 4), which is highly correlated with perceptions of the race at the time of the survey (late October).

5. Appendix B shows additional model specifications that provide near-identical estimates.

status and registration location. We identified students' campus states by merging NSLVE with the Integrated Postsecondary Education Data System and identified students' home states using their residential zip codes (see app. C).

We focus on full-time out-of-state undergraduate students who have SSNs, are over age 18 (at the time of the election), and have an active voting status.⁶ We further omit students from Washington, DC.⁷ This leaves 573,890 observations for 2012 and 651,393 for 2016. The outcome in our analysis is a binary variable coded 1 if students register in their campus state and 0 if they register in their home state. The key explanatory variable, REC, is calculated identically as above using the final projected vote share from fivethirtyeight.com (app. H.2 shows our results hold using actual election outcomes). On average, students seem to move from competitive states to noncompetitive states and vice versa in equal numbers. The mean value of REC in 2012 was -0.46 ($SD = 12.41$), and the mean value for 2016 was 0.38 ($SD = 11.77$). Note that we standardize REC below to facilitate convergence.

We include controls for demographic traits (race, ethnicity, gender), state characteristics (indicators for neighboring states), and college characteristics (public or private).⁸ We also control for potential state-dyadic-level confounding by including the difference in per capita campaign disbursements and the difference in per capita campaign contributions between home states and campus states. Other dyadic controls include the relative timing of the presidential primaries, to account for strategic registration in early primary states, and the relative cost of voting, to account for the relative difficulty of registering.

We test our theory using both the 2012 and 2016 samples in separate logistic regression models that (1) include home-region fixed effects, (2) include campus-state fixed effects, (3) include campus-state and home-region fixed effects, (4) include campus state and home-region random effects, and (5) include home-region, campus-state, and home-state random effects. The results, displayed in table 2, all show a positive and statistically reliable relationship between REC and campus state registration. The coefficients range from a low of 0.136 ($SE = 0.028$) for the state fixed effects model in 2016 to a high of 0.513 ($SE = 0.067$) for the pooled model in that same election. On average, the evidence in favor of our theory is slightly stronger and more stable for 2012 relative to 2016.

These are again quite meaningful results. The average marginal effect for a 2 standard deviation change in REC is 3.9% in 2012 and 2.3% in 2016. This is quite large given the roughly 10% base rate of campus-state voting in this sample. In 2012, in model 5, the predicted probability of campus-state registration for individuals at the 25th percentile value of REC is 7.8%. This rises to 12.2% for the individuals in the 75th percentile.⁹ As an explicit example, again using model 5 in 2012, students from West Virginia attending college in Pennsylvania are expected to be 65% more likely to register in their campus state relative to a student from Pennsylvania attending college in West Virginia (12.7% vs. 7.7%).

DISCUSSION AND CONCLUSION

In this article, we test the theory that OSCS are strategic about where they register. We argue that students are more likely to register in their campus state when the presidential election there is perceived to be more competitive (relative to their home state) and find consistent and robust evidence in favor of this theory using both a novel survey and a large-scale administrative data set. In all, we believe these results represent the most comprehensive test to date of this hypothesis.

In the appendixes, we provide several additional robustness tests including a multilevel model that tests whether this relationship replicates in each state. We find that it largely does, although future studies might seek to investigate the cases where the pattern fails in order to identify scope conditions. We reanalyze the NSLVE data but limit the sample to first-time voters to see whether our result is simply a function of early voting experiences in each state. We also conduct an analysis using only students attending college in neighboring states under the assumption that costs of registering in each location are relatively low. Our results hold with the exception of models 2 and 3 for 2016 in the neighboring states analysis.

Despite these strengths, our findings of course have limitations. First, OSCS may not be representative of the general populace. Additional work is needed to know to what degree a general population responds to similar incentives. Second, as with all observational studies there is always a concern that there is some lurking causally prior confounder. However, the advantage of this question and population is that the nature of the strategic situation OSCS face is, as far as we can tell, exogenous to many plausible confounds (see app. F). The correlation between expected campus state competitiveness and registration rates in the NSLVE data set (including in-state

6. Inactive status can mean either that the voter's address is not verified or that the voter did not participate in any elections in the past few years. Since we are interested in students' registration location, we focus on students' active voting status rather than turnout.

7. Appendix D.4 shows that all results hold when including DC.

8. Appendix C lists data sources and descriptive statistics for all measures.

9. All predicted probabilities here are calculated while holding all other variables at their observed value for each individual. See app. D.2.

Table 2. Registration Choice of US Undergraduates

	Logistic			Generalized Linear Mixed Effects	
	Pooled (1)	Fixed (CampusST) (2)	Fixed (Region, CampusST) (3)	Random (CampusST) (4)	Random (CampusST, HomeST) (5)
2012:					
REC (scaled)	.366 (.046)	.218 (.022)	.209 (.021)	.210 (.008)	.348 (.071)
Constant	−1.400 (.123)	−.619 (.067)	−.579 (.123)	−1.616 (.100)	−1.534 (.125)
AIC	367,612.4	358,824.8	358,133.1	358,414.6	356,237.5
2016:					
REC (scaled)	.513 (.067)	.136 (.028)	.139 (.035)	.142 (.010)	.378 (.069)
Constant	−1.366 (.143)	−.970 (.101)	−1.058 (.133)	−1.628 (.096)	−1.687 (.099)
AIC	393,563.5	382,778.6	380,601	380,887.5	376,705.9
Demographics	✓	✓	✓	✓	✓
Campaign activity	✓	✓	✓	✓	✓
Other controls	✓	✓	✓	✓	✓
Region fixed effects	✓		✓		
Campus state fixed effects		✓	✓		
Regions random effects				✓	✓
Campus state random effects				✓	✓
Home state random effects					✓

Note. Dependent variable: register in campus state. Fixed effects models include clustered standard errors (clustering on campus state). Demographic variables include white, Hispanic, black, and male. Campaign activity variables are relative per capita campaign disbursements and relative per capita contributions. Other control variables are private schools, neighboring state, relative primary election dates, and relative registration difficulty. All coefficients are statistically significant at the $p < .05$ level. AIC = Akaike information criterion. N (2012) = 563,719; N (2016) = 635,436.

students) is 0.022 for 2012 and -0.043 for 2016, suggesting our results are not driven by differential registration rates. Our survey shows that REC is orthogonal to standard individual-level predictors of participation as well as other forms of participation (app. F.1). Further, the design allows us to statistically account for other state-level confounders.

Any remaining confounding must come at the state dyadic level. For instance, one could argue that the relative level of campaign activity in students' home and campus states matters to registration decisions even after accounting for the absolute level of activity in their campus state. While possible, we argue it is unlikely to fully account for this relationship. First, our measures of relative campaign activity (relative per capita campaign expenditures and per capita contributions) are only weakly correlated with REC and often in the wrong direction (app. F.3). Indeed, when we control for these variables in our NSLVE analyses above, our main result remains unchanged, and these potential dyadic-level confounds are

only significant and correctly signed in four out of 20 models despite our large samples (app. C). Further, we fit additional models that simultaneously control for home-state and campus-state factors. Model 3 includes home-region fixed effects while model 5 includes home-state and campus-state random effects. Finally, appendix D.5 shows a model where we include campus-state fixed effects and home-state battleground status, and our conclusion remains unchanged for 2012.

Finally, we can turn to what OSCS themselves report about their registration decisions. In appendix E we provide evidence from a small convenience sample from a university in Missouri in 2016, where OSCS were asked to explain their registration location decisions. In this sample, 61.8% of OSCS indicated that REC influenced their decision, and 48% identified REC as their primary consideration. Further, REC is a stronger predictor of registration location for students who indicated it was a factor.

Finally, although our analysis shows that OSCS do respond to REC as our theory suggests, it does not support overblown claims about undue election influence. (Nor is it clear that students voting in their campus state is normatively undesirable.) In 2016, Hillary Clinton defeated Donald Trump by fewer than 3,000 votes in New Hampshire, making it one of the closest states. According to the NSLVE data set, 13,311 politically active OSCS attended 10 NSLVE schools in New Hampshire. However, only 2,183 registered in New Hampshire. In North Carolina, Governor Pat McCrory lost to Roy Cooper by around 10,000 votes in 2016. However, of the over 17,396 OSCS attending the 21 NSLVE-participating schools only 1,935 registered in North Carolina. Thus, even in the unlikely circumstance that all of the OSCS who registered in their campus state (*a*) voted and (*b*) chose the same candidate, they would not have been pivotal (see app. I). OSCS can be influential, but only for extremely close races. In our view, politicians worried about college students' votes might do better to focus on earning the support of the tens of thousands of registered in-state students rather than disenfranchising the rest.

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